

1.2 Measurements & Errors

Question Paper

Course	CIEA Level Physics
Section	1. Physical Quantities & Units
Topic	1.2 Measurements & Errors
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The density of the material of a coil of thin wire is to be found.

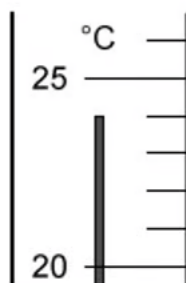
Which set of instruments could be used to do this most accurately?

- A. Metre rule, protractor, spring balance
- B. Micrometer, metre rule, top-pan balance
- C. Stopwatch, newton-meter, vernier calipers
- D. Tape measure, vernier calipers, lever balance

[1 mark]

Question 2

The diagram shows part of a thermometer.



What is the correct reading on the thermometer and the uncertainty in this reading?

	Reading/°C	Uncertainty in reading/°C
A	24	± 1
B	24	± 0.1
C	24.0	± 0.2
D	24.0	± 0.5

[1 mark]

Question 3

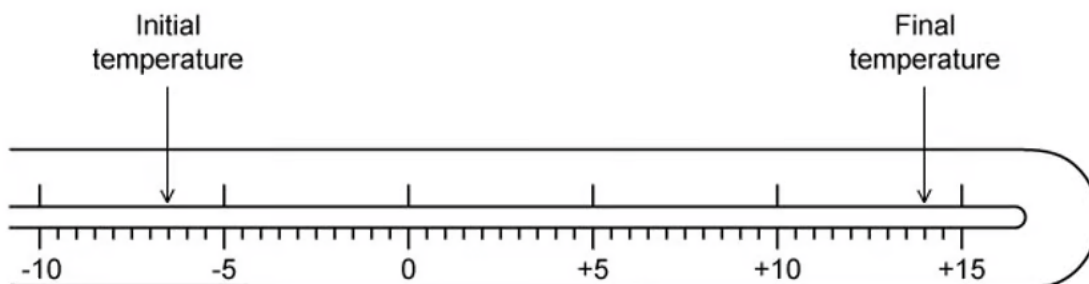
In an experiment to determine the acceleration of free fall using a falling body, what would lead to a value that is too large?

- A. air resistance
- B. dimensions of the body are too large
- C. measured distance longer than true distance
- D. measured time longer than true-time

[1 mark]

Question 4

The diagram shows the stem of a Celsius thermometer marked to show initial and final temperature values.



What is the temperature change expressed to an appropriate number of significant figures?

- A. 14°C
- B. 20.5°C
- C. 21°C
- D. 22.0°C

[1 mark]

Question 5

A metre rule is used to measure the length of a piece of wire. It is found to be 70 cm long to the nearest millimetre.

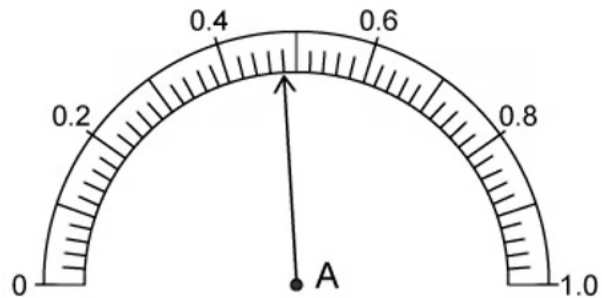
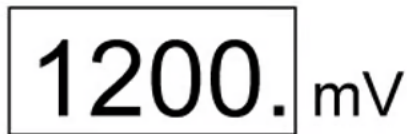
How should this result be recorded in a table of results?

- A. 0.7 m
- B. 0.70 m
- C. 0.700 m
- D. 0.7000 m

[1 mark]

Question 6

The diagrams show digital voltmeter and analogue ammeter readings from a circuit in which electrical heating is occurring.



What is the electrical power of the heater?

- A. 0.53 W
- B. 0.58 W
- C. 530 W
- D. 580 W

[1 mark]

Question 7

An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

- A. The height of fall and the time of fall
- B. The height of fall and the weight of the body
- C. The mass of the body and the height of fall
- D. The mass of the body and the time of fall

[1 mark]

Question 8

A micrometer is used to measure the diameters of two cylinders

diameter of first cylinder = 12.78 ± 0.02 mm

diameter of second cylinder = 16.24 ± 0.03 mm

The difference in the diameters is calculated.

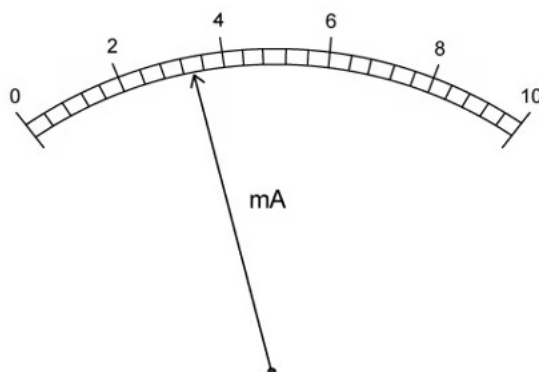
What is the uncertainty in this difference?

- A. ± 0.01 mm
- B. ± 0.02 mm
- C. ± 0.03 mm
- D. ± 0.05 mm

[1 mark]

Question 9

What is the reading shown on this milliammeter?



- A. 2.35 mA
- B. 2.7 mA
- C. 3.4 mA
- D. 3.7 mA

[1 mark]

Question 10

A student carries out a series of determinations of the acceleration of free fall g . The table shows the results.

g / ms^{-2}
4.91
4.89
4.88
4.90
4.93
4.92

What can be said about this experiment?

- A. it is accurate and precise
- B. it is accurate but not precise
- C. it is not accurate and not precise
- D. it is not accurate but is precise

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Density is equal to $\frac{\text{mass}}{\text{volume}}$
- The volume of a wire is equal to $\pi r^2 h$ which means we will have to measure the diameter and the length of the wire
- You would use a micrometer to measure small (< 2.5 cm) diameters that can fit within the 'jaws' of the screw-gauge. Micrometers are accurate to within a hundredth of a millimetre, so this would be the most suitable option to measure the diameter of the wire
- You would use a meter rule to measure medium lengths from 0 to 1 m. Metre rulers have an accuracy of 0.1 cm, so this would be the best option to measure the length of the wire
- A top-pan balance is the best option to measure the mass of the wire as it can measure mass to an accuracy of 0.001 g

A is incorrect because protractors are used to measure angles, so would not be necessary in this case

C is incorrect because stopwatches are used to measure time and newton-meters are used to measure force, neither of which are relevant to density

D is incorrect because even though all of these instruments would be suitable to use, vernier calipers are less accurate than micrometers and a lever balance is much less accurate than a top-pan balance

2

The correct answer is **D** because:

- The uncertainty of a measuring instrument is estimated as plus or minus half the smallest scale division.
- For the thermometer in the question, each mark represents 1.0°C , so the uncertainty is $\pm 0.5^{\circ}\text{C}$
- The reading from this thermometer is 24.0°C , so the result can be written as $24.0 \pm 0.5^{\circ}\text{C}$

A & B are incorrect because the readings are not quoted to the correct precision, and the uncertainties are incorrect

C is incorrect because even though the precision of the reading is correct, the uncertainty is too low

3

The correct answer is **C** because:

- To determine the acceleration of free fall, we will need to consider the equation
 - $s = ut + \frac{1}{2}at^2$
 - We can simplify this to $h = \frac{1}{2}gt^2$ or $g = \frac{2h}{t^2}$ since the falling body will initially be at rest, so $u = 0$ (h = height, g = acceleration of free fall, t = time)
- If the measured distance, h , was measured to be longer than the true distance, this would increase the measured value of g as the two quantities are directly proportional

A is incorrect because air resistance can be treated as negligible in this experiment, but even if we did factor it in, air resistance would increase the time which would lead to a decreased value of g

B is incorrect because the dimensions of the body are not relevant to the experiment

D is incorrect because if the time were measured to be longer than the true

time, this would lead to a decreased value of g since it is inversely proportional to the square of the time (inverse square law)

4

The correct answer is **B** because:

- The thermometer is divided into divisions of 0.5°C
 - The initial temperature is -6.5°C
 - The final temperature is 14.0°C
- Final temperature - initial temperature = $14.0 - (-6.5) = 20.5^{\circ}\text{C}$

5

The correct answer is **C** because:

- The metre rule can measure to the nearest millimetre so the answer must be recorded to 3 significant figures. This is because:
 - A metre rule has an accuracy of $1\text{ mm} = 0.1\text{ cm} = 0.001\text{ m}$
 - So the measurement of 70 cm must be recorded to the same level of accuracy = $70.0\text{ cm} = 0.700\text{ m}$

6

The correct answer is **B** because:

- The reading on the voltmeter is $1200\text{ mV} = 1.2\text{ V}$
- Each division represents 0.02 A so the reading on the ammeter is 0.48 A
- Electrical power is given by
 - $P = VI$
- Power = $1.2 \times 0.48 = 0.576\text{ W} = 0.58\text{ W}$ (2 s. f.)

A is incorrect because this is the answer you get if you read the ammeter scale incorrectly as $I = 0.44\text{ A}$

C is incorrect because this is the answer you get if you use $I = 0.44\text{ A}$ and if

you don't convert the voltage from mV to V

D is incorrect because this is the answer you get if you don't convert the voltage from mV to V and use $V = 1200 \text{ V}$

7

The correct answer is **A** because:

- To determine the measurements needed to calculate the acceleration of free fall, we will need to consider the equation
 - $s = ut + \frac{1}{2}at^2$
 - We can simplify this to $h = \frac{1}{2}gt^2$ since the falling body will initially be at rest, i.e. $u = 0$ (h = height, g = acceleration of free fall, t = time)
- Rearranging the equation above we can see $g = \frac{2h}{t^2}$ so the only measurements required are the height and time of the fall

8

The correct answer is **D** because:

- If we are adding or subtracting quantities then we simply add the absolute uncertainties together
 - $0.02 + 0.03 = 0.05$
- So the uncertainty in the difference in diameters is $\pm 0.05 \text{ mm}$

9

The correct answer is **C** because:

- Each division represents 0.4 mA
- The arrow is halfway between the third and fourth division so the reading is calculated as follows:
 - $2 + 0.4 + 0.4 + 0.4 + 0.2 = 3.4 \text{ mA}$

10

The correct answer is **D** because:

- Precise measurements are consistent, they fluctuate slightly about a mean value
- The measurements are, however, consistent so they are precise
- The true value of g is 9.81 m s^{-2} so the values are not accurate
- The measurements are, however, consistent so they are precise